

**BURLINGTON, WI, USA**WMO: **722059**Lat: **42.690N**Lon: **88.304W**Elev: **237**StdP: **98.51**Time Zone: **-6.00 (NAC)**Period: **04-19**WBAN: **04866****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -21.9      | -18.8 | -26.3                         | 0.4 | -21.1 | -23.2 | 0.5 | -18.1 | 11.5                  | -2.5 | 10.5 | -5.5 | 2.4                   | 280  | 0.536 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.2                   | 32.0            | 23.2 | 30.1 | 22.0 | 28.6 | 21.5 | 24.8                | 29.1 | 23.7 | 28.0 | 22.8 | 26.9 | 4.4                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.5                            | 18.8 | 27.2 | 22.5 | 17.7 | 26.1 | 21.4 | 16.5 | 25.0 | 76.5          | 29.1 | 72.1 | 28.1 | 68.5 | 26.9 | 37.2              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>9.8</b>    | <b>8.5</b> | <b>7.7</b> | <b>-25.5</b>               | <b>34.9</b> | <b>4.2</b>         | <b>3.8</b> | <b>-28.5</b>                                       | <b>37.7</b> | <b>-31.0</b> | <b>39.9</b> | <b>-33.3</b> | <b>42.1</b> | <b>-36.4</b> | <b>44.8</b> |    |    |
| (5)               |            |            | <b>-25.7</b>               | <b>27.4</b> | <b>4.2</b>         | <b>3.1</b> | <b>-28.7</b>                                       | <b>29.6</b> | <b>-31.2</b> | <b>31.4</b> | <b>-33.5</b> | <b>33.2</b> | <b>-36.5</b> | <b>35.4</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.2    | -6.2  | -5.3  | 1.3   | 7.7   | 14.1  | 19.2  | 21.2  | 20.1  | 16.5  | 9.8   | 2.6   | -3.3 |
|                                                                                  | DBStd       | 11.03  | 6.71  | 6.59  | 6.26  | 5.01  | 4.91  | 3.82  | 3.17  | 3.02  | 4.49  | 5.15  | 5.78  | 5.99 |
|                                                                                  | HDD10.0     | 2043   | 503   | 428   | 281   | 103   | 17    | 0     | 0     | 0     | 3     | 67    | 228   | 412  |
|                                                                                  | HDD18.3     | 3982   | 762   | 662   | 529   | 322   | 149   | 33    | 10    | 17    | 87    | 269   | 472   | 670  |
|                                                                                  | CDD10.0     | 1389   | 0     | 0     | 10    | 33    | 143   | 278   | 346   | 312   | 199   | 61    | 7     | 1    |
|                                                                                  | CDD18.3     | 287    | 0     | 0     | 1     | 2     | 18    | 61    | 97    | 71    | 33    | 5     | 0     | 0    |
|                                                                                  | CDH23.3     | 2906   | 0     | 0     | 6     | 33    | 233   | 603   | 952   | 662   | 359   | 56    | 1     | 0    |
|                                                                                  | CDH26.7     | 841    | 0     | 0     | 1     | 4     | 63    | 192   | 284   | 176   | 108   | 13    | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.1    | 3.8   | 3.7   | 3.6   | 3.9   | 3.2   | 2.6   | 2.2   | 1.9   | 2.3   | 3.0   | 3.4   | 3.5  |
| Precipitation                                                                    | PrecAvg     | 892    | 49    | 43    | 58    | 85    | 104   | 117   | 85    | 94    | 80    | 65    | 64    | 51   |
|                                                                                  | PrecMax     | 1108   | 101   | 115   | 116   | 171   | 267   | 211   | 195   | 295   | 169   | 171   | 132   | 114  |
|                                                                                  | PrecMin     | 657    | 9     | 3     | 16    | 36    | 27    | 23    | 27    | 33    | 12    | 10    | 12    | 19   |
|                                                                                  | PrecStd     | 122    | 24    | 29    | 31    | 35    | 57    | 56    | 40    | 56    | 42    | 44    | 36    | 26   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.6  | 15.1  | 23.8  | 27.1  | 31.1  | 33.7  | 34.1  | 32.4  | 32.5  | 28.3  | 20.8  | 13.6 |
|                                                                                  |             | MCWB   | 24.6  | 11.7  | 16.1  | 16.1  | 21.4  | 23.3  | 25.1  | 22.6  | 22.2  | 21.7  | 14.5  | 11.3 |
|                                                                                  | 2%          | DB     | 8.4   | 9.1   | 17.5  | 23.5  | 28.4  | 31.0  | 31.3  | 30.5  | 29.8  | 24.3  | 16.5  | 10.0 |
|                                                                                  |             | MCWB   | 7.9   | 6.2   | 13.0  | 14.9  | 19.4  | 22.4  | 23.8  | 22.2  | 21.6  | 17.4  | 11.7  | 8.7  |
|                                                                                  | 5%          | DB     | 5.0   | 5.2   | 14.0  | 20.1  | 26.1  | 28.8  | 29.3  | 28.5  | 27.5  | 21.5  | 13.7  | 7.0  |
|                                                                                  |             | MCWB   | 3.4   | 2.9   | 10.1  | 13.6  | 18.1  | 21.2  | 22.3  | 21.7  | 19.5  | 16.0  | 10.4  | 5.7  |
|                                                                                  | 10%         | DB     | 2.5   | 2.9   | 10.9  | 17.2  | 23.4  | 27.1  | 27.9  | 27.2  | 25.2  | 18.4  | 11.3  | 4.7  |
|                                                                                  |             | MCWB   | 1.7   | 1.4   | 7.9   | 11.5  | 17.0  | 20.1  | 21.5  | 20.7  | 18.9  | 14.7  | 8.4   | 3.4  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.5  | 11.8  | 16.8  | 18.8  | 23.3  | 25.7  | 27.2  | 25.3  | 24.1  | 22.9  | 15.5  | 12.6 |
|                                                                                  |             | MCDB   | 22.5  | 13.6  | 22.3  | 23.9  | 28.0  | 30.5  | 31.0  | 29.9  | 29.6  | 27.5  | 19.3  | 13.3 |
|                                                                                  | 2%          | WB     | 7.9   | 6.7   | 14.1  | 16.4  | 20.9  | 23.9  | 25.0  | 23.9  | 22.6  | 19.7  | 12.7  | 8.3  |
|                                                                                  |             | MCDB   | 8.1   | 8.8   | 17.6  | 20.8  | 26.0  | 28.7  | 28.9  | 27.9  | 27.4  | 22.3  | 15.7  | 9.4  |
|                                                                                  | 5%          | WB     | 3.7   | 3.5   | 11.0  | 14.2  | 19.5  | 22.4  | 23.9  | 23.0  | 21.4  | 17.3  | 10.8  | 5.9  |
|                                                                                  |             | MCDB   | 4.7   | 4.9   | 12.9  | 19.8  | 24.2  | 27.0  | 28.0  | 26.9  | 25.7  | 19.5  | 13.6  | 6.9  |
|                                                                                  | 10%         | WB     | 1.7   | 1.7   | 7.4   | 11.9  | 17.9  | 21.2  | 22.9  | 21.9  | 20.1  | 15.0  | 8.6   | 3.3  |
|                                                                                  |             | MCDB   | 2.4   | 3.0   | 10.1  | 16.8  | 21.8  | 25.4  | 26.8  | 25.5  | 24.1  | 18.1  | 11.2  | 4.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.1   | 8.9   | 9.9   | 11.5  | 12.5  | 12.0  | 12.2  | 12.3  | 13.2  | 11.5  | 9.6   | 7.7  |
|                                                                                  |             | MCDBR  | 9.6   | 11.4  | 14.7  | 17.3  | 16.5  | 14.0  | 13.3  | 13.9  | 15.6  | 16.0  | 13.4  | 9.4  |
|                                                                                  |             | MCWBR  | 8.3   | 9.1   | 9.9   | 10.0  | 8.6   | 6.8   | 6.6   | 6.8   | 7.7   | 9.2   | 9.2   | 8.1  |
|                                                                                  | 5% WB       | MCDBR  | 8.5   | 10.9  | 13.0  | 15.6  | 13.9  | 12.0  | 12.0  | 11.6  | 13.5  | 13.0  | 12.2  | 9.1  |
|                                                                                  |             | MCWBR  | 7.6   | 9.0   | 9.7   | 9.8   | 8.3   | 6.8   | 7.0   | 6.5   | 7.1   | 9.2   | 9.2   | 8.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.287  | 0.310 | 0.341 | 0.391 | 0.413 | 0.434 | 0.430 | 0.419 | 0.389 | 0.344 | 0.305 | 0.283 |      |
|                                                                                  | taud        | 2.391  | 2.332 | 2.374 | 2.258 | 2.247 | 2.231 | 2.241 | 2.268 | 2.332 | 2.461 | 2.507 | 2.442 |      |
|                                                                                  | Ebn at Noon | 855    | 888   | 902   | 875   | 862   | 842   | 841   | 839   | 838   | 842   | 833   | 829   |      |
|                                                                                  | Edh at Noon | 78     | 98    | 107   | 130   | 135   | 138   | 135   | 127   | 111   | 86    | 70    | 68    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.60   | 2.56  | 3.63  | 4.61  | 5.34  | 6.06  | 6.13  | 5.31  | 4.35  | 2.80  | 1.78  | 1.29  |      |
|                                                                                  | RadStd      | 0.22   | 0.30  | 0.34  | 0.38  | 0.40  | 0.41  | 0.30  | 0.33  | 0.36  | 0.33  | 0.16  | 0.12  |      |